BMVC 2026 Pivot Brief — Engagement VLM Paper

Date: 2026-05-06 · Deadlines: Abstract 22 May, Paper 29 May (no extensions)

Where we are

CV4Edu (CVPR Workshop): Accepted to non-archival track. Doesn't bar later peer-reviewed submission.

CV4Edu archival track: Rejected. Reviewer cLFD (confidence 5) flagged: single-frame protocol invalidates temporal task; N=300 sampling biased; SCB scene-level reformulation not comparable to prior work; GPT-4o L2 fallback corrupts metrics; "pure benchmarking without methodological innovation is weak."

Plan: Resubmit a substantially strengthened version to BMVC 2026.

Original BMVC plan (Framing A — "Diagnose-and-Adapt")

Reviewer signal said benchmark-only is dead. The plan kept the diagnostic half and added an adaptation half: linear probing, class-prior calibration, multi-frame inference, ordinal CLIP. Goal: *"the signal is in the encoder, the language readout is broken — here's how to fix it."*

The paper hinged on one experiment: **subject-disjoint linear probe on CLIP features.**

- κ_quad $\geq$ 0.25 → encoder has signal → Framing A works
- κ_quad < 0.15 → encoder lacks signal → Framing A fails, pivot needed

Decision: drop SCB; run on DAiSEE only (cleaner ordinal task, official subject-disjoint splits, comparable to prior literature).

What we ran

Linear probes on the **full 1,784-clip DAiSEE test set**, trained on the full official train split (5,358 clips, subject-disjoint). Three independent variations to rule out single-encoder artifacts:

Probe	κ_quad (95% CI)
CLIP ViT-B/32, single frame, LogReg (balanced)	0.055 [0.014, 0.091]
CLIP ViT-B/32, single frame, Ridge (ordinal)	0.098 [0.062, 0.134]
CLIP ViT-B/32, mean of 3 frames, LogReg	0.101 [0.068, 0.133]
CLIP ViT-B/32, mean of 3 frames, Ridge	0.103 [0.061, 0.140]
DINOv2 ViT-B/14, single frame, LogReg	0.107 [0.072, 0.143]
DINOv2 ViT-B/14, single frame, Ridge	0.090 [0.042, 0.133]
Best zero-shot (LLaVA P3) — for reference	~0.10

What this tells us

A $\kappa \approx$ 0.10 ceiling holds across three orthogonal axes:

(1) Encoder family — CLIP (vision-language) ≈ DINOv2 (vision-only). (2) Temporal context — 1 frame ≈ 3 frames. (3) Readout — categorical ≈ ordinal regression.

The best probe is statistically tied with the best zero-shot prompt. None reach 0.15. Framing A's premise (signal in encoder, broken readout) is **decisively rejected**.

Proposed new direction (Pivot 1 — "The Engagement Ceiling")

The negative result is stronger than the original benchmark because it pre-empts every reviewer attack:

> *"Frozen vision features — whether CLIP, DINOv2, or zero-shot VLMs; whether single-frame or temporal; whether read out via prompts or supervised heads — encode an upper bound of $\kappa \approx 0.10$ on individual classroom engagement. The bottleneck is representational, not in the prompt or language readout."*

Survives: "you didn't try adaptation" (we did), "single-frame is the problem" (multi-frame also fails), "CLIP-specific quirk" (DINOv2 also fails).

Implies: the field needs face/AU-aware encoders or end-to-end fine-tuning, not better prompts. Direct contribution to BMVC's explainable-AI track or Brave New Ideas track.

Remaining work (~21 days)

(1) Full 1,784-clip zero-shot reruns with bootstrap CIs (CLIP, LLaVA, GPT-4o, Qwen2.5-VL). (2) Multi-frame Qwen2.5-VL inference (closes temporal-handicap critique reviewer-side). (3) Class-prior calibration on existing logits (sanity check). (4) Drop SCB; switch to BMVC template; rewrite intro around the ceiling thesis. (5) Ethics paragraph (DAiSEE consent / IRB).

Questions for you

(1) On board with Pivot 1, or push on additional probes (face-cropped CLIP, deeper probe) before locking framing?

(2) Title preference: *"The Engagement Ceiling: Why Frozen Vision Features Cannot Recover Individual Classroom Engagement"* — or alternatives?

(3) Anything to keep from the SCB scene-level work as a side analysis, or drop entirely?